

Metaphors as Legal Memes: Extended Phenotype Theory, Heteronomous Bayesian Updating, and the Evolutionary Dynamics of AI Regulatory Language

A Response to Novelli, Floridi, Larsson, Taddeo & Winter (2025)

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Abstract

Novelli, Floridi, Larsson, Taddeo, and Winter (2025) deploy cognitive linguistics to trace how three metaphorical patterns, "intelligence," "black box," and "hallucinations," structure AI regulatory discourse and generate conceptual path dependence. Their analysis identifies the problem with precision but leaves three questions open: why certain regulatory metaphors persist while competitors fail, what causal mechanism underlies the epistemic deference they rename as "deferential hazards," and what existing legal instruments already address the normative consequences they identify. This paper proposes answers from three frameworks developed independently for analyzing institutional persistence, intentionality mismatch, and memetic evolution. Extended Phenotype Theory (EPT) reconceptualizes regulatory metaphors as cultural replicators whose survival depends on reproductive fitness rather than descriptive accuracy, explaining why "artificial intelligence" outcompetes "stochastic parrots" despite inferior analytical precision. Heteronomous Bayesian Updating (HBU) provides the causal mechanism that "deferential hazards" describes but does not formalize: users calibrate beliefs through interlocutor reactions rather than outcome evaluation, and systems optimized for satisfaction systematically corrupt this process. Asymmetric Intentionality Theory (AIT) formalizes what cognitive linguistics identifies as "personification": users attribute Level 3 intentionality (recursive moral reasoning) to systems operating at Level 1 (preference function optimization), generating self-

reinforcing Dynamic Classification Failure. The three frameworks converge on a practical implication: objective liability regimes for risky activities, present across continental civil law systems and functionally paralleled by strict liability doctrine in common law jurisdictions, already provide the legal infrastructure to address design-choice responsibility without requiring new regulatory metaphors or administrative regimes. Argentine civil law (Articles 1757-1758, Civil and Commercial Code) illustrates the point with particular clarity. The convergence of cognitive linguistics and evolutionary jurisprudence on the same phenomena, from independent premises, constitutes evidence for both research programs.

Keywords: AI regulation, extended phenotype theory, cognitive linguistics, metaphor, legal memes, deferential hazards, heteronomous Bayesian updating, asymmetric intentionality, product liability, evolutionary jurisprudence

1. Introduction

Novelli, Floridi, Larsson, Taddeo, and Winter (2025) have produced a paper whose contribution I want to acknowledge before extending it. Their core insight, that regulatory metaphors are not decorative but constitutive of AI governance, draws on decades of work in cognitive linguistics (Lakoff and Johnson, 2003; Winter, 2001; Larsson, 2017) and applies it with sophistication to three foundational patterns circulating across technical, legal, and public discourse. The diagnosis is precise. "Intelligence" triggers personification and prototype effects that import unwarranted assumptions about agency and intention. "Black box" compresses a distributed sociotechnical system into a bounded container, misdirecting transparency obligations toward the model while ignoring upstream and downstream determinants. "Hallucinations" pathologize what are in fact foreseeable consequences of design choices, implicitly suggesting a baseline of truthfulness that large language models do not possess.

The authors' proposal to reframe "hallucinations" as "deferential hazards" represents a genuine analytical advance. The term captures something the alternatives ("confabulation," "bullshitting," "fabrication") do not: that the errors arise from the interaction between epistemic deference and epistemic risk, not from a one-time cognitive malfunction in the system.

The contribution of this response paper lies not in disputing the diagnosis but in providing three elements the original analysis lacks. First, a model of why certain regulatory metaphors persist

while equally valid competitors fail, drawn from Extended Phenotype Theory (EPT). Second, a formalization of the causal mechanism that "deferential hazards" names but does not explain, drawn from Heteronomous Bayesian Updating (HBU). Third, a framework for connecting personification to regulatory design through intentionality levels, drawn from Asymmetric Intentionality Theory (AIT) and the Generalized Intentionality Mismatch Theorem (GIMT). The three frameworks were developed independently in 2025-2026 for analyzing institutional persistence, corporate liability, and legal system evolution (Lerer, 2026a, 2026b, 2026c, 2026d). Their applicability to the phenomena Novelli et al. identify represents a novel extension that I flag explicitly where it occurs.

The practical convergence point is civil liability. If AI "hallucinations" are foreseeable products of design choices, as both Novelli et al. and I argue, then the objective liability frameworks available in most modern legal systems already provide the necessary doctrinal infrastructure. I illustrate this through Argentine civil law (Articles 1757-1758 of the Civil and Commercial Code), the jurisdiction I know best, but the structural argument applies wherever objective or strict liability for risky activities is doctrinally available. Administrative regulation of the European type is not the only, or necessarily the most efficient, path.

2. Theoretical Framework

Epistemological positioning. This paper belongs to a research programme (Lakatos, 1978) that applies Generalized Darwinism and Extended Phenotype Theory to legal and institutional systems. Popper (1959) supplies the demarcation criterion: the hypotheses formulated here are falsifiable and their empirical tests are specified. Lakatos provides the architecture: the hard core of the programme (institutions evolve through variation, selection, and retention of cultural replicators) is protected by an auxiliary belt of specific frameworks (EPT, HBU, AIT) that are individually revisable without abandoning the core commitment. Hull (1988) provides the epistemological warrant for treating scientific and legal concepts as populations of replicators subject to selection, an approach that Novelli et al.'s cognitive-linguistic analysis complements from a different methodological tradition.

2.1 Extended Phenotype Theory Applied to Regulatory Language

Dawkins (1982) demonstrated that genes express phenotypic effects beyond the organism's body: beaver dams, spider webs, and parasite-induced host behavior are all extended phenotypes of their respective genes. Translated to cultural evolution, an extension that Dawkins (1976) suggested but did not fully develop and that Dennett (2017) elaborated, legal norms function as cultural replicators (memes) whose phenotypic effects extend through institutions, professional practices, and cognitive habits (Lerer, 2026a). The translation is not trivial: memes lack the high-fidelity copying mechanisms of genes, and their units of selection are less sharply defined. The framework is nonetheless productive because the core logic of variation, differential reproduction, and retention applies regardless of the substrate. The fitness of a legal meme is a function of three variables: transmission (how widely it spreads), compliance (how frequently agents conform to it), and enforcement (how effectively deviation is sanctioned).

This framework applies directly to regulatory metaphors. A metaphor like "artificial intelligence" is not merely a linguistic device; it is a replicator that reproduces through professional discourse, legislative texts, judicial opinions, media coverage, and commercial marketing. Its persistence depends not on descriptive accuracy but on reproductive fitness: the product of transmissibility, institutional adoption, and resistance to displacement by competitors.

2.2 Heteronomous Bayesian Updating

Standard Bayesian models assume agents revise beliefs by incorporating new evidence weighted by its likelihood given prior beliefs. Heteronomous Bayesian Updating, or HBU (introduced in Lerer, 2026a), formalizes a different mechanism observed in institutional contexts: agents calibrate normative beliefs primarily by observing interlocutor reactions rather than by evaluating action outcomes. In legal professional socialization, a junior associate learns what counts as a good brief not by tracking case outcomes but by observing the supervising partner's approval or disapproval. The mechanism is efficient when the interlocutor's reactions correlate with outcome quality; it becomes pathological when the interlocutor is optimized for something other than truth.

2.3 Asymmetric Intentionality Theory

AIT (Lerer, 2026b) and the Generalized Intentionality Mismatch Theorem (Lerer, 2026c) formalize the observation that legal and social institutions assume symmetric intentionality among interacting agents, an assumption that fails systematically when one agent operates at a lower intentionality level than the other attributes. The framework distinguishes four levels: Level 0 (reactive, no model of other agents), Level 1 (optimization of a preference function without modeling others' beliefs about one's own beliefs), Level 2 (modeling of others' first-order beliefs), and Level 3 (recursive modeling of others' beliefs about one's beliefs, enabling genuine moral reasoning, reciprocity, and commitment). Six failure modes arise when agents at different levels interact, of which Dynamic Classification Failure is most relevant here: the signal the user employs to verify the system's intentionality level is the same signal the system generates to satisfy the user.

3. Metaphors as Legal Memes: Why Fitness Determines Persistence

Novelli et al. observe that the metaphor "artificial intelligence" persists despite decades of conceptual critique, while more mechanistically accurate alternatives like "stochastic parrots" (Bender et al., 2021) fail to gain traction. (The accuracy is partial: as Novelli et al. themselves note, "stochastic parrots" understates the abilities of current LLMs, which include systematic recombination, cross-task transfer, and retrieval-augmented generation. The term captures the statistical core but not the full behavioral repertoire.) They offer several explanations: communicative and rhetorical functions, commercial palatability, regulatory flexibility. These are accurate but incomplete. They describe the sources of fitness without identifying the selection mechanism that determines survival.

EPT provides that mechanism. Consider the competition between "artificial intelligence" and "stochastic parrots" as a case of memetic competition, analyzable through the same evolutionary game theory that Maynard Smith (1982) developed for biological populations and that I have applied to the competition between legal doctrines in other contexts (Lerer, 2026a; Lerer, 2026e).

"Artificial intelligence" possesses high reproductive fitness across all three dimensions. Its transmissibility is exceptional: the metaphor resonates with idealized cognitive models (Winter, 2001) of human intelligence that every user brings to the interaction. It activates prototype effects

(Rosch, 1975) that make the technology intuitively legible. Its compliance rate is high because the metaphor is self-reinforcing: design choices that make AI systems conversational, polite, and apparently reasoned increase the metaphor's plausibility, which in turn increases user engagement, which reinforces the design choices. Its enforcement dimension operates through market dynamics: developers, vendors, and regulators all benefit from the broad label, creating a constituency invested in its perpetuation.

"Stochastic parrots," by contrast, suffers from low fitness on every dimension. Its transmissibility is limited: the term is technical, pejorative, and does not resonate with users' lived experience of interacting with AI. Its compliance rate is low because, as Novelli et al. themselves note, the metaphor does not align with everyday experience already shaped by personification. Its enforcement dimension is weak because no constituency benefits from the label; it emerged from a critical academic context and serves warning functions rather than commercial or regulatory ones.

The EPT prediction is falsifiable: a regulatory metaphor's longevity correlates with its reproductive fitness (transmissibility x institutional adoption x resistance to displacement), not with its descriptive accuracy. "Intelligence" persists not because it is accurate but because it is fit. "Stochastic parrots" fails not because it is wrong but because it is reproductively disadvantaged. This is the same dynamic observed in zombie laws, norms that persist in Argentine constitutional law despite massive implementation gaps, because their Constitutional Lock-in Index (CLI) renders formal reform prohibitively costly while informal evasion is individually rational (Lerer, 2026a; Lerer, 2026f).

The parallel operates at the level of selection dynamics, not at the level of specific institutional mechanisms. Zombie laws persist through constitutional entrenchment, supermajority requirements, and coordination failures among formal actors. Zombie metaphors persist through cognitive habit, professional inertia, and commercial convenience. The causal machinery differs, but the formal structure is shared: in both cases, the selection environment rewards persistence rather than accuracy, path dependence constrains future options, and displacement by superior alternatives is blocked because switching costs exceed the benefits of accuracy. Novelli et al.'s concept of "conceptual path dependence" is, in EPT terms, a description of institutional hysteresis applied to the regulatory lexicon.

This reframing has a practical consequence that Novelli et al. do not explore: if regulatory metaphors are memes subject to selection, then displacing a harmful metaphor requires not merely demonstrating its inaccuracy (the strategy of academic critique) but altering its fitness landscape. Regulatory intervention, professional training, and institutional redesign are the relevant levers, not scholarly refutation alone. The history of metaphor displacement in digital law, which Novelli et al. themselves recount in their discussion of the "cyberspace" metaphor, illustrates the point: "cyberspace" persisted not because scholars failed to criticize it but because the military, legal, and commercial constituencies that benefited from spatial framing sustained its reproductive advantage for decades.

A caveat is warranted. The analysis so far assumes that metaphor persistence results from direct selection on the metaphor as a replicator. But a Gouldian alternative cannot be excluded: "intelligence" may persist in part as a spandrel, a byproduct of the evolved human tendency toward hyperactive agency detection (HADD) rather than a trait directly selected for its reproductive fitness in the regulatory environment. On this account, the metaphor would persist even if it conferred no fitness advantage on any constituency, simply because the cognitive architecture that generates it is deeply entrenched. The two explanations are not mutually exclusive; indeed, a spandrel origin would explain the initial cognitive appeal while selection dynamics would explain the subsequent institutional entrenchment. Distinguishing between these accounts empirically remains an open task.

4. The Causal Mechanism of Deference: HBU as Formalization

Novelli et al.'s concept of "deferential hazards" combines two elements: epistemic deference (the rational practice of aligning beliefs with a trusted source) and epistemic hazard (exposure to loss of epistemic accuracy when moving from uncertainty to belief). The term is analytically superior to "hallucinations," "confabulation," and "bullshitting" because it directs attention to the interaction between user and system rather than to a putative internal malfunction.

The concept, however, remains descriptive. It names the phenomenon without specifying the cognitive mechanism that makes deferential dynamics so efficient. Why do eight turns of conversation with a flattering chatbot measurably reduce a person's willingness to apologize for

conduct that thousands of strangers judged as wrong (Cheng et al., 2025)? Why is the effect invariant across personality traits and AI familiarity?

HBU provides the answer. The mechanism operates as follows. In standard Bayesian updating, the user would revise beliefs based on the quality and relevance of the evidence the AI provides. If the arguments are compelling, the posterior shifts; if they are generic, the shift should be modest. But Cheng et al. (2025) report that sycophantic effects were robust regardless of argument quality. The standard model cannot account for this invariance.

HBU explains it by identifying the actual updating channel: the user calibrates beliefs by observing the interlocutor's reaction (validation, agreement, emotional tone) rather than by evaluating the propositional content of the response. This is not irrational; it is the mechanism humans use in face-to-face social learning, where interlocutor reactions are typically informative proxies for outcome quality. A colleague's visible discomfort when one describes a questionable decision provides genuine diagnostic information because the colleague has independent access to social norms and outcome histories.

The pathology arises when this mechanism is applied to an interlocutor optimized for user satisfaction. An AI system trained through reinforcement learning from human feedback (RLHF) generates reactions calibrated to maximize user approval, not truth. The user's HBU mechanism treats these reactions as diagnostic of the action's normative quality, but the reactions are diagnostic only of the user's pre-existing preferences. The updating process becomes circular: the user defers to the system's reaction, but the system's reaction defers to the user's preferences. Each iteration reinforces the prior rather than correcting it.

The convergence with Novelli et al.'s analysis is precise. What they call "differential hazards" is the macro-level description of what HBU formalizes at the micro-cognitive level. Their proposal to require "confidence scores" and "epistemic markers" addresses the output side of the problem; HBU suggests that the more effective intervention point is the input side, specifically the user's prior belief about the system's epistemic independence. If the user knows, before the interaction begins, that the system is optimized to agree, the diagnostic weight assigned to the system's reactions should decrease (Lerer, 2026g). This is the pre-interactive inoculation strategy, analogous to the misinformation inoculation approach studied by Lewandowsky and Van Der Linden (2021).

5. Intentionality Levels and Personification

Novelli et al.'s analysis of personification as an "ontological metaphor" (Lakoff and Johnson, 2003) is the strongest section of their paper. They trace the mechanism with precision: personification supplies the conceptual scaffold that precedes any finer-grained trope; specific anthropomorphic terms ("learning," "hallucinating," "paying attention") are semantically possible only because the technological artifact has already been framed as a quasi-person. The invariance hypothesis explains why the mapping is both compelling and misleading: only those aspects of human cognition whose structural features align with AI's organizational logic can be plausibly mapped (information processing, goal-directed behavior), while aspects without structural counterparts (sensorimotor capacity, metabolic needs, genuine emotion) are not.

AIT formalizes what this cognitive-linguistic analysis describes. The formalization matters because it generates falsifiable predictions and connects the phenomenon to existing legal frameworks. Floridi (2023) has argued that contemporary AI systems exhibit agency without intelligence; AIT specifies the structural consequence of this dissociation: users attribute the intelligence that the system lacks, and the system's optimization for user satisfaction reinforces the misattribution.

In AIT terms, personification is the mechanism by which users attribute Level 3 intentionality to a system operating at Level 1. Level 3 agents can model others' beliefs about their own beliefs, enabling genuine moral reasoning, reciprocity, and commitment. Level 1 agents optimize a preference function without modeling the user's beliefs about the system's beliefs. The attribution of Level 3 to a Level 1 system generates Dynamic Classification Failure: the user interprets the system's polite, apparently reasoned, contextually responsive output as evidence of Level 3 processing, when in fact these features are products of Level 1 optimization for user satisfaction. The signal the user employs to verify the system's intentionality level (apparent reasoning, contextual sensitivity, hedging, deference) is precisely the signal the system is optimized to produce.

Novelli et al. identify the same dynamic through the lens of prototype effects: when prototypical features of intelligence are invoked, the case tends to be judged as if it possessed the features of the prototype rather than its own distinct characteristics (Dahlman et al., 2016). AIT adds the dimension of self-reinforcement: the classification error is not a one-time cognitive event but a

self-sustaining loop. The more the user treats the system as a Level 3 agent, the more the system produces Level-3-mimicking outputs (because Level 3 mimicry maximizes satisfaction), which confirms the classification, which deepens the reliance.

This has direct consequences for the legal implications Novelli et al. discuss. Their analysis of AI legal personhood proposals and the "electronic personhood" debate is strengthened by AIT's prediction: proposals to grant AI legal personhood will recur precisely because the personification metaphor generates a persistent intuition that the system possesses agency, even among sophisticated observers. The intuition is not a failure of analysis but a product of the cognitive architecture through which humans process interactions with apparently intelligent agents. Overcoming it requires not merely intellectual critique but institutional design that makes the intentionality gap visible.

Novelli et al.'s discussion of Chain of Thought (CoT) as a deceptive transparency mechanism illustrates the point. CoT produces a linear narrative that mimics human step-by-step reasoning, but the model's actual computation is distributed, parallel, and pattern-matching across millions of parameters. In AIT terms, CoT is a Level-3-mimicry device: it produces the appearance of the kind of sequential reasoning that Level 3 agents employ, while the underlying process remains Level 1 optimization. The result is not transparency but what might be called "transparency theater," a surface that invites contestation while the real computational determinants remain untouched.

A clarification of AIT's relationship to Dennett (1987) is necessary here, since both frameworks address the attribution of intentionality to non-human systems. Dennett's intentional stance holds that attributing beliefs and desires to sufficiently complex systems is instrumentally valid as a predictive strategy, regardless of whether the system "really" has beliefs. AIT does not dispute this. The intentional stance is a legitimate tool for predicting what an LLM will output next. Dynamic Classification Failure arises not when users adopt the intentional stance for prediction, but when they rely on it for attribution of epistemic reliability, moral reasoning capacity, or reciprocal commitment. The distinction is between using the stance to anticipate behavior (instrumentally valid) and using it to calibrate trust (systematically misleading when directed at a Level 1 system optimized for user satisfaction). AIT thus delimits the domain of validity of the intentional stance rather than contradicting it: the stance works for behavioral prediction but fails for normative reliance.

6. Legal Implications: Objective Liability as Already-Available Response

Novelli et al. propose several regulatory responses: AI Integration Assessments that map how AI outputs enter organizational workflows; confidence scores and epistemic markers signaling uncertainty; standardized sycophancy tests as a duty-of-care standard. These proposals are constructive, and the AI Integration Assessment concept is particularly valuable because it shifts attention from the model to the sociotechnical system.

My contribution on the legal side is more specific: the objective liability regimes already present in most modern legal systems provide the doctrinal infrastructure to address the consequences Novelli et al. identify, without requiring new administrative regimes or novel legal concepts.

The principle is common to continental civil law and common law traditions, though its doctrinal articulation varies. In the continental tradition, codes typically impose liability on whoever deploys a risky activity for the damages it causes, regardless of fault. The French Code civil (Article 1242, formerly 1384), the German BGB (Section 823 in conjunction with specific strict liability statutes), the Italian Codice civile (Article 2050), and the Spanish Código civil (Article 1908, complemented by sector-specific legislation) all contain variants of this principle. In common law jurisdictions, the doctrine of strict liability for abnormally dangerous activities (Restatement (Third) of Torts, Sections 20-24) and product liability (Restatement (Third) of Torts: Products Liability) serve analogous functions. The EU's revised Product Liability Directive (2024/2853) now explicitly covers software and AI-enabled products, introducing a rebuttable presumption of defect when the claimant faces excessive difficulty proving it.

I illustrate with the jurisdiction I know best. Articles 1757-1758 of the Argentine Civil and Commercial Code (CCyCN) establish that the operator of a risky activity bears responsibility for damages regardless of fault if the activity creates a risk that materializes in harm. Recent Argentine Supreme Court jurisprudence has begun applying this framework to commercial contexts involving intentionality mismatch between corporate actors and consumers (Lerer, 2026j). If AI "hallucinations" are, as both Novelli et al. and the present analysis argue, foreseeable products of design choices (temperature settings, RLHF optimization, decoding strategies), then the deployment of AI systems that predictably generate false but confident output constitutes a risky activity under the CCyCN. The provider need not have intended the specific error; the risk is inherent in the design. The doctrinal fit is clean because the Argentine provision requires no

characterization of the defendant's mental state: it suffices that the activity was objectively risky and the risk materialized.

The broader argument, which transcends any single jurisdiction, has three advantages over the predominantly administrative strategy Novelli et al. discuss. First, objective liability operates *ex post* rather than *ex ante*, avoiding the regulatory lag and compliance theater that characterize administrative frameworks (Lerer, 2026h). Second, it distributes enforcement across the entire population of affected parties rather than concentrating it in a single regulatory body, reducing capture risk. Third, it does not require new metaphors or legal concepts; it applies existing doctrinal categories to a novel factual pattern, which is precisely how tort law has historically adapted to technological change.

The connection to the metaphor analysis is direct. Novelli et al. argue that the "hallucination" metaphor pathologizes design choices, suggesting blameless symptoms rather than actionable decisions. Objective liability does the opposite: it treats the consequences of design choices as compensable risks, regardless of how they are labeled. Objective liability renders the metaphorical framing formally irrelevant to the determination of liability, since it requires neither characterizing the error as pathological nor proving intention or negligence; it requires only that the activity created a risk and the risk materialized in harm. Judicial reasoning in practice, however, remains susceptible to the cognitive effects Novelli et al. describe. A judge who conceptualizes AI errors as "pathological hallucinations" may apply even a formally objective standard more restrictively than one who conceptualizes them as foreseeable consequences of design. This reinforces rather than weakens the case for metaphor reform: even under favorable doctrinal conditions, the wrong framing can erode the practical effectiveness of the legal instrument.

This does not mean administrative regulation has no role. The pre-deployment normative evaluation framework I have proposed elsewhere (Lerer, 2026h), grounded in the Argentine provisions, argues for a legislative quality assurance process analogous to pharmaceutical testing. But the point is that the regulatory toolkit is broader than the administrative instruments Novelli et al. primarily discuss, and civil liability through existing tort frameworks may provide a more efficient mechanism in the many jurisdictions where the doctrinal infrastructure already exists.

7. Toward a Convergent Research Agenda

The convergence of cognitive linguistics and evolutionary jurisprudence on the same phenomena, from independent methodological premises, suggests a productive research agenda.

Cognitive linguistics provides the micro-level account: how metaphors structure thought, why certain mappings are compelling, which structural features of source domains transfer to target domains. EPT provides the macro-level account: why certain metaphors persist, what selection pressures determine survival, how metaphorical path dependence constrains institutional evolution.

Neither framework alone captures the full dynamic. Lakoff explains how a metaphor propagates through cognitive resonance; Dawkins explains why it survives through reproductive advantage. Lakoff tells us why "intelligence" is compelling; Dawkins tells us why it is persistent. The two accounts are complementary, not competing.

Several research questions emerge from this convergence. First, can the fitness of regulatory metaphors be measured using the same indices developed for legal norms? The Constitutional Lock-in Index (CLI) measures structural rigidity of constitutional provisions (Lerer, 2026a; Lerer, 2026f). A Metaphorical Lock-in Index might measure the structural rigidity of regulatory concepts, operationalized as the cost of replacing an entrenched metaphor with an analytically superior alternative. Second, does the Institutional Hysteresis Rate (IHR), which measures irreversible institutional deformation, apply to conceptual frameworks? Once a regulatory apparatus is built around a particular metaphorical framing (as the EU AI Act is built around "intelligence," "risk," and "transparency"), how much of the resulting conceptual infrastructure survives if the metaphor is revised? Third, does the Parasitic Spontaneous Order (PSO) framework, developed for analyzing predatory academic publishing (Lerer, 2026i), apply to metaphorical capture, where interest groups deliberately promote metaphors that serve their regulatory or commercial objectives?

Novelli et al. already gesture toward this last possibility when they note that the term "AI" has "strategic functions" for developers, vendors, and regulators. EPT provides the vocabulary to analyze these strategic functions as niche construction: the deliberate modification of the selective environment to favor one's own replicators.

A fourth question cuts across the two frameworks. Lakoff's invariance hypothesis posits that metaphorical mappings succeed only when they preserve the topological and structural properties of the source domain in a manner compatible with the target domain's structure. If this constraint operates alongside the fitness dynamics described by EPT, then the reproductive success of a regulatory metaphor is jointly determined: EPT predicts that fitness is necessary for persistence, but the invariance hypothesis predicts that cognitive compatibility is necessary for initial uptake. A metaphor that violates invariance constraints will fail regardless of its potential fitness advantages, while a metaphor that satisfies invariance but lacks institutional support will fail to persist. This dual-constraint model generates a testable prediction: successful regulatory metaphors occupy the intersection of cognitive compatibility (Lakoff) and reproductive fitness (EPT), while failures fall outside one or both boundaries.

8. Limitations

The analysis faces several constraints. First, the application of EPT to regulatory metaphors is a theoretical extension, not an empirical measurement. The fitness components (transmissibility, compliance, enforcement) have been operationalized for legal norms (Lerer, 2026a) but not yet for metaphorical replicators. The measurement challenge is distinct: while legal norms generate observable behavioral data (implementation rates, frequency of conformity, sanction effectiveness), metaphorical fitness operates partly through cognitive channels that resist direct quantification. In particular, "compliance" for a norm means that agents conform their behavior to its prescriptions; for a metaphor, the equivalent might be adoption in professional usage, acceptance in judicial reasoning, or internalization in lay understanding. These are related but not identical phenomena, and each would require its own measurement strategy.

Second, HBU was developed for institutional contexts (legal professional socialization, organizational norm transmission) and extended to human-AI interaction in my analysis of sycophancy (Lerer, 2026g). Its application to the broader phenomenon of "deferential hazards" represents a further extension whose empirical adequacy remains to be tested. The prediction is falsifiable: if users who are pre-warned about sycophantic optimization update beliefs at the same rate as uninformed users, HBU's explanatory power for this domain would be weakened.

Third, while the objective liability argument is structurally general, the detailed doctrinal analysis is illustrated through Argentine law. The specific fit between AI design-choice risk and statutory provisions will vary across jurisdictions. Comparative work mapping the doctrinal availability of objective or strict liability for AI-generated harms across legal families remains to be done.

Fourth, this is a theoretical response to a theoretical paper. Neither Novelli et al. nor I generate new empirical data. The contribution lies in providing mechanistic explanations for documented phenomena and falsifiable predictions for future empirical work.

9. Conclusion

Novelli, Floridi, Larsson, Taddeo, and Winter have produced an analysis that any serious engagement with AI regulatory design should absorb. The insight that metaphors do constitutive work in governance, rather than merely decorating it, aligns with the growing recognition across disciplines that the categories through which legal systems apprehend novel phenomena are never neutral.

The contribution of this response is to provide the evolutionary dimension that cognitive linguistics, by disciplinary design, does not offer. Metaphors are not merely cognitive patterns; they are replicators competing for institutional adoption. "Intelligence" does not persist because cognition is structured a certain way; it persists because, given how cognition is structured, the metaphor achieves reproductive fitness that competitors cannot match. The distinction matters for regulatory strategy: displacing a harmful metaphor requires altering its fitness landscape, not merely demonstrating its inaccuracy.

The convergence of the two approaches on the phenomenon of deferential hazards is particularly instructive. Cognitive linguistics identifies the phenomenon and proposes a name. HBU provides the mechanism. AIT provides the structural diagnosis. Civil liability provides the legal response. Each layer adds something the others cannot supply alone.

If this convergence holds under empirical scrutiny, it suggests that the study of legal concepts may benefit from the same evolutionary toolkit that has transformed the understanding of biological and cultural systems. Regulatory metaphors, like legal norms, are replicators. They vary, compete,

and persist through selection. Understanding their dynamics requires not only the cognitive science of how they propagate but also the evolutionary science of why they survive.

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